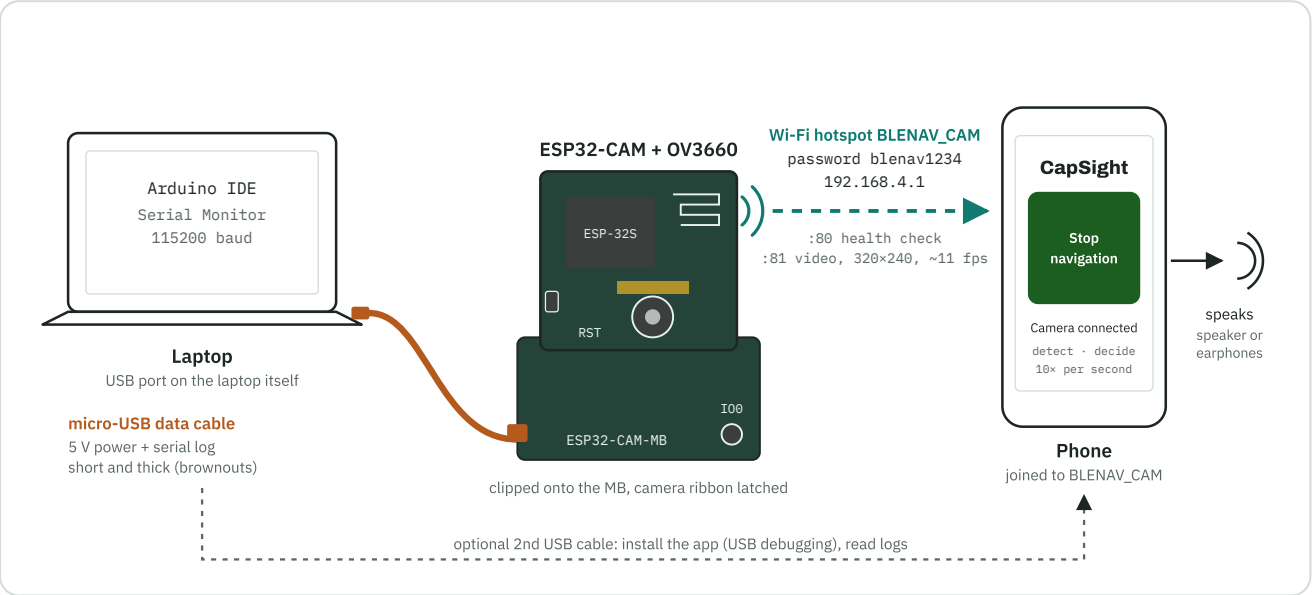
CAPSIGHT · PHASE B BENCH TEST · NO BATTERY

Run CapSight with the ESP32-CAM on the MB, powered from the laptop

The ESP32-CAM sits on its ESP32-CAM-MB programmer, and one micro-USB cable from the laptop powers it. The board makes its own Wi-Fi hotspot. The phone joins that hotspot, runs the detection and speaks the warnings. The laptop only supplies power and shows the board's log. It does not need to join the camera's Wi-Fi.



USB power + serial (laptop → MB) ---- Wi-Fi radio (board → phone)

- - - optional, for installing the app

One USB cable powers the camera and carries its log to the laptop. All video goes over the board's own Wi-Fi straight to the phone, which does every step from detection to speech.

WI-FI NAME	PASSWORD	CAMERA ADDRESS	SERIAL BAUD	EXPECTED VIDEO
BLENAV_CAM	blenav1234	http://192.168.4.1	115200	320×240 · ~11 fps

Steps

1 Assemble the stack and plug it into the laptop

- Lift the camera connector latch on the ESP32-CAM, push the OV3660 ribbon fully in, close the latch.
- Clip the ESP32-CAM onto the ESP32-CAM-MB. All pins go in, and the MB's micro-USB port faces away from the camera.
- Connect the MB to a USB port **on the laptop itself** with a short **data** cable. Avoid hubs and long thin cables: the board draws up to about 500 mA while streaming.

2

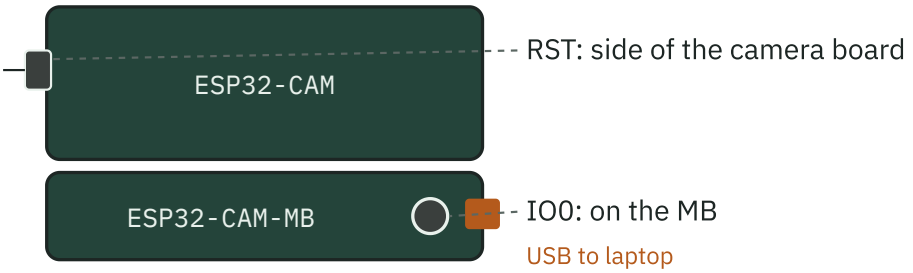
FIRST TIME, OR AFTER A FIRMWARE CHANGE

Flash the firmware from Arduino IDE

Open `firmware/esp32_cam_ap/esp32_cam_ap.ino` and set:

```
Board        AI Thinker ESP32-CAM
Partition    Huge APP (3MB No OTA)
Upload speed 115200
Port         the MB's port (e.g. /dev/ttyUSB0 or /dev/ttyACM0)
```

Click Upload. The MB switches the board into flash mode by itself. If you see `Failed to connect to ESP32`, hold **IO0** on the MB, tap **RST** on the ESP32-CAM, release IO0, then click Upload again. After the upload finishes, tap **RST** once so the new firmware starts.



3

Check that the board started

In Arduino IDE open Tools → Serial Monitor at **115200** baud, then tap RST.

> **You should see:** the BLENAV_CAM banner with `Join WiFi : BLENAV_CAM` and the stream address. `[cam] INIT FAILED` means the ribbon is not seated: unplug the cable, reseat the ribbon, plug it back in.

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OPTIONAL

Check the stream from the laptop

```
bash tools/stream_check.sh
```

It moves the laptop onto BLENAV_CAM for about 30 seconds, records 10 s of video, saves the first frame to `/tmp/capsight_stream_check/first_frame.jpg`, then switches back to your normal Wi-Fi. The camera serves **one viewer at a time**, so close any browser tab showing `192.168.4.1` before the phone connects.

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Join the phone to the camera's Wi-Fi

Phone Settings → Wi-Fi → **BLENAV_CAM** → password **blenav1234**. When Android says the network has no internet, choose **Stay connected** (or "Keep Wi-Fi connection"). Mobile data can stay on, because the app sends its camera traffic over this Wi-Fi network only.

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FIRST TIME, OR AFTER AN APP CHANGE

Install the app on the phone

With a second USB cable to the phone (USB debugging on), from the project folder:

```
export
JAVA_HOME=/var/lib/flatpak/app/com.google.AndroidStudio/current/ac
tive/files/extra/jbr
./gradlew --no-daemon installDebug
```

Or send `app/build/outputs/apk/debug/app-debug.apk` over WhatsApp as a document and tap it. Use the USB cable, not wireless debugging, because the phone leaves your home Wi-Fi when it joins BLENAV_CAM.

7

Point the app at the real camera and start

- Open CapSight → **Debug view** → Frame source: **ESP32-CAM over Wi-Fi**.
- Tap **Use real cap (192.168.4.1)** and leave **Bind to Wi-Fi network** on.
- Go back and tap the big green **Start navigation** button.

> **You should hear** "Navigation started". The debug view shows **Camera connected** and `cam ~11 fps 320x240`, with `FOV 54° (cap setting)`.

8

Try it

- Aim the camera at a chair. At about 2.5 m you should hear "Chair ahead. Move left/right." At about 1.2 m you should hear "Stop. Chair ahead." and feel a vibration.
- Pull the USB cable out of the MB. Within about 2 s the phone says **"Camera disconnected. Please stop."** Plug it back in: the board reboots in a few seconds and the phone says "Camera connected." by itself (allow up to about 10 s).
- To finish, tap **Stop navigation** or use Stop in the notification. You should hear "Navigation stopped".

If something goes wrong

WHAT YOU SEE	LIKELY CAUSE	FIX
Board reboots every few seconds, stream keeps dropping	Brownout: the USB port or cable cannot supply enough current	Use a shorter, thicker cable directly on a laptop port, not a hub
BLENAV_CAM missing from the phone's Wi-Fi list	Firmware not running, or still in flash mode	Tap RST and check the Serial Monitor for the banner
Upload fails with "Failed to connect to ESP32"	Board did not enter flash mode	Hold IO0, tap RST, release IO0, upload again
App stays on "Connecting to camera"	Phone not on BLENAV_CAM, address changed, or Bind to Wi-Fi off	Rejoin BLENAV_CAM; in Debug view tap "Use real cap" and turn Bind on
Connected but 0 fps, or a frozen picture	Camera clock raised above 8 MHz (it jams this board's antenna), or a browser tab is holding the stream	Close the browser tab; reset the clock with `http://192.168.4.1/control?xclk=8` or tap RST
Phone drops BLENAV_CAM after a minute	Android prefers networks with internet	Forget and rejoin, choose "Stay connected"; turn off "Switch to mobile data" if offered

This is the bench setup for testing. On the cap, a power bank replaces the laptop, the MB stays at home and the board runs by itself. CapSight supplements the white cane; it does not replace it.